Sales Data Transformation and Data Modeling

Objective:

This project focuses on analyzing sales data using Power BI, DAX measures, and interactive visualizations. The dataset was cleaned, transformed, and modeled to generate meaningful business insights. Various DAX calculations such as Total Sales, Average Sales, Profit Analysis, Order Count, and Year-to-Date (YTD) Sales were created to evaluate business performance. Interactive dashboards and visual reports were developed to track sales trends, customer behavior, regional performance, and key performance indicators (KPIs), enabling data-driven decision-making.

Dataset Description:

The dataset contains the following columns.

Column description:

Column Name	Description
Order Id	Order ID of the Product
Order Date	Order Date of the Product
Customer Name	Customer Name of the Product
State	State Name of the Customer
City	City Name of the Customer
Amount	Price of the Product
Profit	Profit of the Product
Quantity	Quantity of the Product
Category	Category of the Product
Sub-Category	Sub-Category of the Product
Month of Order Date	Order Date of the Sales
Target	Target for the Sales
Profit Margin	Newly Calculated the Profit Margin

Formulas Used:

Calculated Columns

Create a Calculated Column for 'Category Type':

```
Category Type = 'Order Details (2)'[Category]&"-"&'Order Details
(2)'[Sub-Category]
```

Calculate Revenue per Order in Order Details Table:

```
Revenue = 'Order Details (2)'[Amount]*'Order Details (2)'[Quantity]
```

Create a Calculated Column to Categorize Sales:

```
Sales Category = IF('Order Details (2)'[Amount]>AVERAGE('Order Details
(2)'[Amount]),"Above Average","Below Average")
```

Calculated Measures

Average Profit in Delhi:

```
Average Profit in Delhi = CALCULATE(AVERAGE('Order Details
(2)'[Profit]),'List of Orders (2)'[City]="Delhi")
```

Order Count:

```
Order Count = COUNT('Order Details (2)'[Order ID])
```

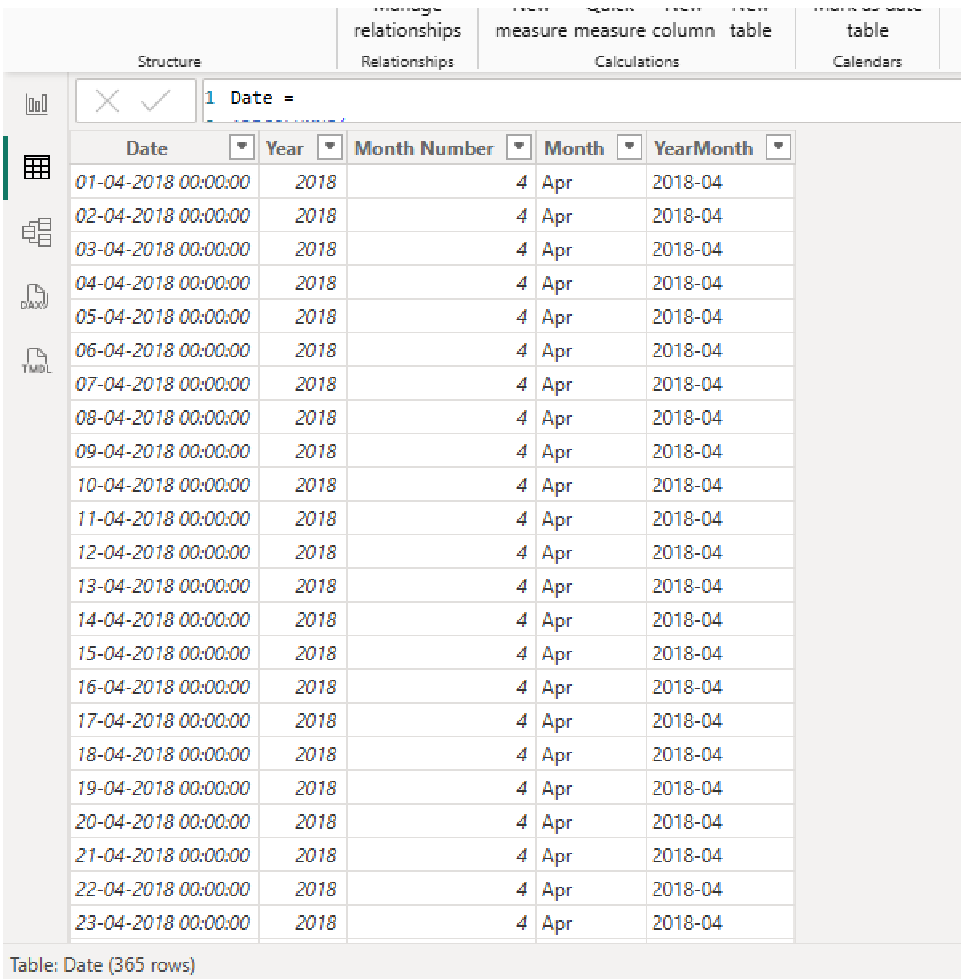
Year-to-Date (YTD) Sale:

```
Year to Date Sales = TOTALYTD(SUM('Order Details
(2)'[Revenue]),'Date'[Date])
```

Year	Year to Date Sales
2018	**₹ 13,10,271**
4	₹ 1,52,827
5	₹ 2,91,594
6	₹ 4,01,609
7	₹ 4,54,753
8	₹ 6,06,267
9	₹ 7,38,984
10	₹ 9,15,647
11	₹ 11,45,432
12	₹ 13,10,271
2019	**₹ 8,36,599**
1	₹ 3,37,229
2	₹ 5,14,849
3	₹ 8,36,599
Total	**₹ 8,36,599**

Calender Date Table Creation

```
Date =
ADDCOLUMNS(
   CALENDAR(DATE(2018,4,1), DATE(2019,3,31)),
   "Year", YEAR([Date]),
   "Month Number", MONTH([Date]),
   "Month", FORMAT([Date],"MMM"),
   "YearMonth", FORMAT([Date],"YYYY-MM")
)
```

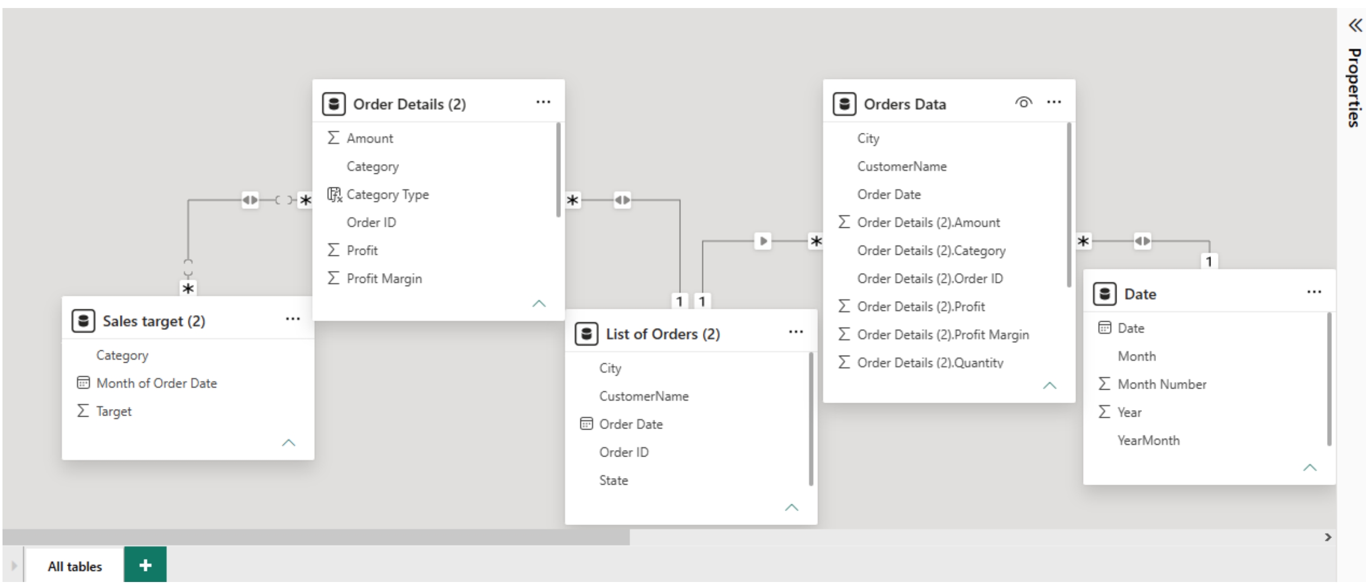


Date	Year	Month Number	Month	YearMonth
01-04-2018 00:00:00	2018	4	Apr	2018-04
02-04-2018 00:00:00	2018	4	Apr	2018-04
03-04-2018 00:00:00	2018	4	Apr	2018-04
04-04-2018 00:00:00	2018	4	Apr	2018-04
05-04-2018 00:00:00	2018	4	Apr	2018-04
06-04-2018 00:00:00	2018	4	Apr	2018-04
07-04-2018 00:00:00	2018	4	Apr	2018-04
08-04-2018 00:00:00	2018	4	Apr	2018-04
09-04-2018 00:00:00	2018	4	Apr	2018-04
10-04-2018 00:00:00	2018	4	Apr	2018-04
11-04-2018 00:00:00	2018	4	Apr	2018-04
12-04-2018 00:00:00	2018	4	Apr	2018-04
13-04-2018 00:00:00	2018	4	Apr	2018-04
14-04-2018 00:00:00	2018	4	Apr	2018-04
15-04-2018 00:00:00	2018	4	Apr	2018-04
16-04-2018 00:00:00	2018	4	Apr	2018-04
17-04-2018 00:00:00	2018	4	Apr	2018-04
18-04-2018 00:00:00	2018	4	Apr	2018-04
19-04-2018 00:00:00	2018	4	Apr	2018-04
20-04-2018 00:00:00	2018	4	Apr	2018-04
21-04-2018 00:00:00	2018	4	Apr	2018-04
22-04-2018 00:00:00	2018	4	Apr	2018-04
23-04-2018 00:00:00	2018	4	Apr	2018-04

Conclusion:

The project demonstrates best practices in DAX calculations and data visualization that support effective business intelligence and decision-making.

Data Modelling:



DashBoard Layout: